

DUNCAN WOOD

Washington, D.C. Metro Area | duncanwoodpro@gmail.com

Portfolio: duncanwoodpro.netlify.app | LinkedIn: linkedin.com/in/duncanwoodpro | GitHub: github.com/Duncan-Wood

SUMMARY

Practical, action-oriented Software Engineer with 2+ years of experience delivering technical solutions in high-ambiguity environments. Proven track record in rapid solution delivery, compressing time-to-value from months to weeks, and stabilizing high-stakes systems. Skilled in Python, SQL, and cloud infrastructure, with a demonstrated ability to sit at the frontline of complex technical problems, troubleshoot integration friction, and translate requirements into reliable, production-grade tools.

TECHNICAL SKILLS

- **Data & Pipeline Engineering:** Python (FastAPI, Flask, SQLAlchemy), SQL (PostgreSQL, BigQuery), ETL/ELT pipelines, Supabase.
- **Cloud & Deployment:** Google Cloud Platform (Cloud Run, Cloud Functions, Cloud Batch), Docker, CI/CD, Sentry, Application Security.
- **AI & Rapid Prototyping:** Vector Embeddings (RAG), LLM Integration, Next.js, React, TypeScript, Ruby on Rails.

EXPERIENCE

Software Engineer — EcoMap Technologies, Inc. | November 2023 – July 2026

- **Platform Ownership & Stabilization:** Stepped into sole engineering ownership of a flagship 70+ customer white-labeled multi-tenant platform during an abrupt organizational transition. Spearheaded the full client-delivery and stabilization lifecycle—managing rapid triage, scoping, and deployment—improving system reliability and reducing customer escalations to <3/month, focusing on long-term platform stability.
- **Rapid 0→1 Value Delivery:** Architected and delivered an internal real-time data-review tool utilizing a Next.js, TypeScript, and Supabase stack to compress data-team verification workflows. Handled the full lifecycle from schema design to production adoption, utilizing parallel and intercepted-route patterns to resolve severe state desynchronization and eliminate operational friction.
- **Technical Execution & On-site Delivery:** Independently designed and deployed two core analytics submodules (Population and Business Formation) on a flagship Ruby on Rails platform. Engineered high-performance BigQuery SQL models, automated monthly data-refresh pipelines via background jobs, and resolved critical customer-reported data-accuracy discrepancies to maintain deep stakeholder trust.
- **AI/LLM Time-to-Value Optimization:** Designed and deployed a reusable Rails-based LLM summarization concern across three analytics modules using structured output schemas. Compressed inference costs by ~5x and enhanced data accuracy by leading a production RAG chatbot's embedding model migration across 27,000+ records in Supabase.
- **Security & Incident Engagement:** Root-caused and remediated a critical production SQL-injection vulnerability on a core Python service. Led a high-priority cross-functional credential rotation response, migrating 4 infrastructure services from a static shared API key to per-consumer independently rotatable secrets with zero client downtime.
- **Reliability & Observability Hardening:** Engineered custom error-triage systems in Sentry across multiple Next.js codebases to filter out high-volume false-positive environmental noise from real production crashes. Improved platform reliability by proactively simulating network and service outages to harden critical endpoints, effectively preventing production crashes during high-traffic periods.
- **Pattern Capture & Cost Reduction:** Conceived and built self-directed search log analysis tooling to map infrastructure value streams. Identified that ~50% of traffic originated from automated bots, executing an upstream fix that drove a tangible reduction in infrastructure spend while capturing clear operational metrics.
- **Cross-functional Analytics:** Acted as primary technical liaison for product stakeholders, utilizing Pandas and Jupyter Notebooks to aggregate cross-platform data, informing strategic roadmap decisions and resolving complex customer-data queries.

Contract Full-Stack Developer — Mighty Crow | January 2025 – November 2025

- **Compliance Architecture:** Designed a rule-based compliance engine using database triggers and functions to automate real-time housing certification state-cascades across regulatory registries, ensuring strict data consistency and accuracy.
- **Microservices:** Independently designed and shipped a HIPAA-compliant Python/Flask document-conversion service on Google Cloud Run, replacing an expensive commercial alternative with a performant, cost-optimized solution including a full automated test suite for unit, integration, and performance coverage.

PROJECTS

- **LinkedIn Web Scraper (Scrapy):** Engineered an automated pipeline to ingest structured market signals for continuous analytical processing, including custom logic to handle dynamic DOM elements and rate-limiting to maintain high-integrity data retrieval.
- **Hemingway Search Engine:** Built a full-stack search engine leveraging Word2Vec embeddings and NumPy arrays for advanced similarity-based document retrieval, automated corpus management, and optimized indexing performance. Engineered high-performance indexing strategy using optimized tokenization, achieving sub-100ms retrieval latency across large-scale document corporuses.

EDUCATION

- **General Assembly** — Software Engineering Immersive (April 2023)
- **University of Central Florida** — B.A., Writing & Rhetoric (May 2020)